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Cellulose-derived carbon aerogels: A novel porous platform for supercapacitor electrodes

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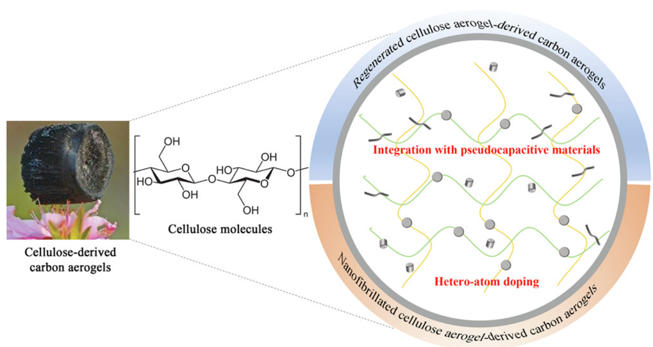
HIGHLIGHTS

- Summarize the preparation strategies of CDCAs from RCAs and NCAs.
- Analyze the role of hetero-atom doping in improving energy storage ability of CDCAs.
- Summarize the synergistic effect of CDCAs and integrated pseudocapacitive materials.
- Unveil the challenges and perspectives of CDCA-based electrodes and devices.

GRAPHICAL ABSTRACT



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ABSTRACT

With the coming of the green chemistry era, the tunable porosity and superb environmental friendliness accompanied by outstanding electrical conductivity of cellulose-derived carbon aerogels (CDCAs) have remarkably promoted the development of high-performance green supercapacitor electrodes. In this review, a detailed overview on the preparation strategies of CDCAs and their precursors (regenerated cellulose aerogels and nanofibrillated cellulose aerogels) are presented. Two crucial approaches for the improvement of energy storage ability, i.e., hetero-atom doping and integration with pseudocapacitive materials, are summarized. Moreover, the role of CDCAs in the energy storage ability is analyzed from the view of structure–response relationship. A comprehensive overview of the fundamental aspects related to the configuration and energy storage mechanism of supercapacitors and the synthesis, structure, and property of cellulose is provided. Finally, this review presents a brief summary, challenges, and perspectives of CDCA-based supercapacitor electrodes and devices.

Contents

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1. Introduction

The rapid consumption of non-renewable resources (like petroleum and coals) has brought serious environmental concerns and energy crisis. Therefore, various renewable energy sources (for instance, wind energy, solar energy, tidal energy, hydroenergy, and geothermal energy) have attracted increasing attention from the scientific community and industrial community [1-9]. In daily life, the most usual application pattern is to transform these renewable energy sources to electric energy with the help of multifarious electric generators [10]. Moreover, the surplus electric energy needs to be effectively stored, hence facilitating the birth and development of energy storage devices [11-13]. Supercapacitors have been extensively considered as one of the quite promising energy storage systems due to their excellent reversibility (90–95% or higher), high power capability, long cycle life (>10^5), and safety of operation [14-17]. Other advantages include fast charging-discharging rate, maintenance-free operation, and wide operation temperature [18]. Supercapacitors are mainly made up of electrodes, separators, and electrolytes. Among them, electrodes play a crucial role in the energy storage performances of supercapacitors [19]. On the basis of working mechanisms, the supercapacitor electrodes can be divided into two categories: electrical double-layer capacitive materials (EDLC, typically like carbon materials) and pseudocapacitive materials (typically like conductive polymers and metallic compounds). For EDLC materials, the energy storage is based on the electrostatic interaction between ions on the large specific surface area of active electrode materials and electrolytes [20]. For the latter, the energy storage relies on the quick and reversible Faradaic redox reactions on the surface and bulk at the vicinity of surface between electrodes and electrolyte ions [21]. Compared to pseudocapacitive materials, EDLC materials usually feature with faster charging-discharging rate and stronger cycling stability as well as more superior rate capability, but smaller specific capacitance and lower energy density. For the sake of remedying these limitations, these two classes of materials are commonly integrated together to develop high-performance supercapacitor electrodes.

Carbon materials, e.g., graphite [22], graphene [23,24], carbon nanotubes (CNTs) [25], carbon fibers [26], carbon aerogels [27], carbon foams [28], and activated carbon [29], are the most usual EDLC materials because of their fascinating merits including variety of form (0D to 3D), ease of processability, tunable porosity, and affluent electrocatalytic active sites for a variety of reactions. For carbon materials, both the specific surface area and pore size are the two most significant parameters in terms of energy storage capacity and power delivery rate [30]. Among various carbon materials, carbon aerogels have own unique merits like large surface area [31], good absorption [32], high porosity [33], hierarchical pore structure [34,35], good high-temperature resistance [36], and low density [37]. Now carbon aerogels-based electrodes have been paid much attention. Carbon aerogels are nanostructured carbons obtained from the carbonization of organic aerogels which are prepared by the sol–gel polycondensation of certain organic monomers. Traditionally, typical organic precursors of carbon aerogels include polyurethane aerogels [38], phenolic aerogels [39], resorcinol–furfural aerogels [40], etc. However, these traditional carbon aerogels suffer from high density (100–800 mg cm^{-3}), fragility, and environmental unfriendliness during the organic reactions [41]. As a result, the greener cellulose-derived carbon aerogels (CDCAs) emerge. Cellulose is the most abundant natural organic polymer on Earth and widely exists in the cell wall of green plants and algae. This polysaccharide ($(C6H10O5)n$) is made up of a linear chain of several hundred to many thousands of $\beta(1 \rightarrow 4)$ linked D-glucose units. For the preparation of CDCAs, the precursors (i.e., cellulose aerogels) are fabricated in prior. There are two kinds of cellulose aerogels (regenerated cellulose aerogels (RCAs) and nanofibrillated cellulose aerogels (NCAs)) that are classified according to their preparation methods [42]. RCAs are prepared through a dissolution- regeneration-freeze drying or supercritical drying process [43,44]. By contrast, the dissolution-regeneration process is replaced with the nanofibrillation of cellulose for NCAs [45]. RCAs are characteristic with stronger and thicker skeleton structure, as compared to NCAs, and thus deliver stronger structural strength and higher density. On the contrary, NCAs display lower density and more superior flexibility as well as stronger adsorption ability because the diameter of the main component (cellulose nanofibrils) is only dozens of nanometers. With the help of high-temperature pyrolysis under the protection of inert gas, both RCAs and NCAs are readily transformed into CDCAs. In supercapacitors, CDCAs not only are directly used to fabricate EDLC electrodes, but can also serve as the substrate or template to assemble pseudocapacitive materials through multifarious physicochemical techniques (like electrodeposition [46], hydrothermal method [47], and gas polymerization [48]). The advantages of CDCAs in supercapacitors mainly include: (1) the large surface area, which

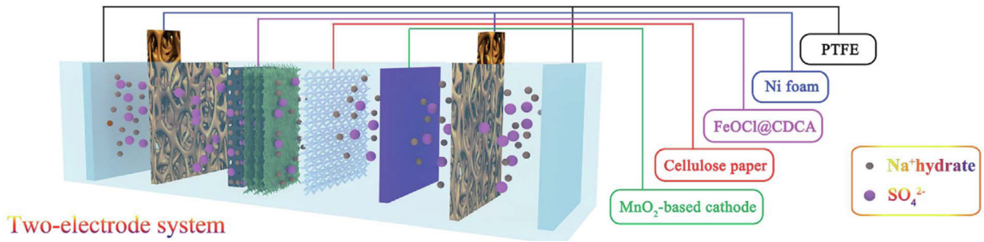

Fig. 1. Configuration of a MnO2//FeOCl@CDCA asymmetric supercapacitor (ASC). (Positive electrode: MnO2; Negative electrode: FeOCl@CDCA; Separator: cellulose paper; Current collector: Ni foam; Packaging material: PTFE plate). (Reprinted with permission from [57]. Copyright 2019 The Royal Society of Chemistry.).

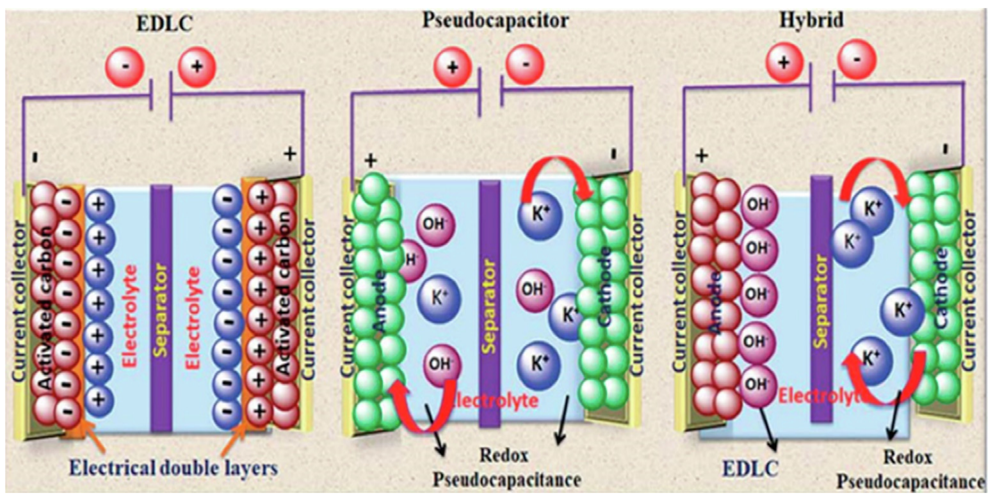

Fig. 2. Charge storage mechanism of different types of supercapacitors. (Reprinted with permission from [60]. Copyright 2019 The Royal Society of Chemistry.).

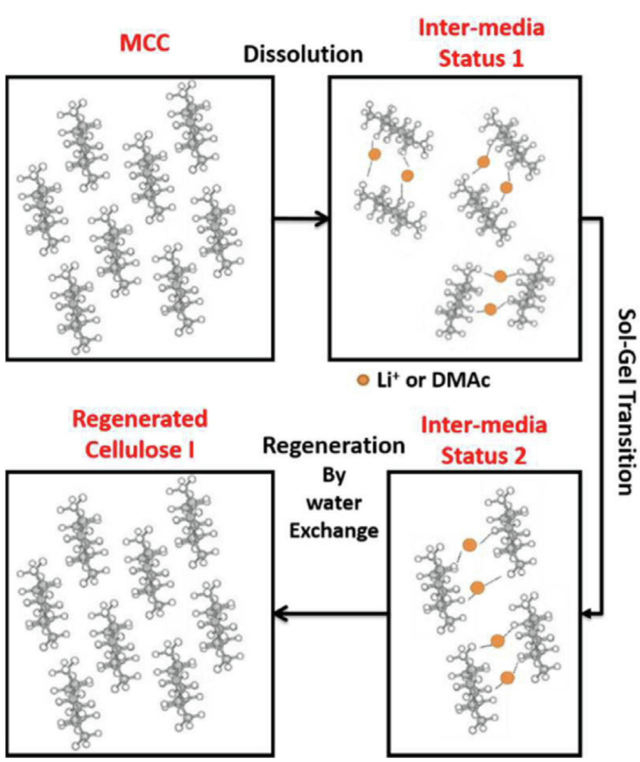

Fig. 3. Regeneration mechanism of cellulose I from its LiCl/DMAc solution. (Reprinted with permission from [72]. Copyright 2017 The Royal Society of Chemistry.).

ranges from hundreds of square meters to thousands of square meters per gram, contributes to increasing the number of electrochemically active sites; (2) the high porosity of CDCAs can serve as the reservoir of electrolyte ions, which shortens the diffusion path and increases the migration efficiency of ions; (3) the high electrical conductivity of CDCAs can act as the high-speed road for electron transferring; (4) the hierarchical open pore structure provides abundant space for the in-situ growth of guest functional materials, resulting in the production of novel CDCA-based composites; (5) good economic effectiveness and excellent environmental benefits.

Recently, some excellent review papers on the preparation, characterization and applications of carbon aerogels have been published [14,49,50]. Nevertheless, most of the reviews do not precisely focus on the carbon aerogels prepared from cellulose but resin-based carbon aerogels [50,51] or pay more attention to graphene or CNTs-based carbon aerogels [52-54]. Also, reviews on the carbon aerogels for supercapacitor applications are rare. The aim of this review is to demonstrate the state-of-the-art technology of CDCAs in the field of supercapacitors. The roles of the unique structure and multi-level pore size distribution of CDCAs in the energy storage property are emphasized. Moreover, the fundamentals including the configurations and working mechanisms of supercapacitors, structure and physicochemical property of cellulose, and preparation technologies of CDCAs are also elaborated. We expect that this summarization can be helpful for the development and innovation of next-generation green supercapacitor electrodes. Finally, this review provides a brief summary, challenges, and perspectives of CDCA-based supercapacitor electrodes.

2. Configuration and energy storage mechanisms of supercapacitors

2.1. Configuration

The configuration of supercapacitors is similar to that of batteries. In a basic form, there are two electrodes (the positive electrode and negative electrode) that are separated by a semipermeable membrane (the separator) for preventing electrical short circuits. For a traditional electrochemical capacitor, it stores electrical energy in the electrical double layer that forms at the interface between an electrolytic solution and an electronic conductor. For pseudocapacitors or hybrid supercapacitors, the charge storage relies on redox reactions. The positive electrode (cathode) is related to reductive chemical reactions that gain electrons from the external circuit. The negative electrode (anode) is associated with oxidative chemical reactions which release electrons into the external circuit. In addition, the separator is a physical barrier between the positive and negative electrodes to prevent electrical shorting. The main forms of separators include porous membranes, flexible hydrogels, and porous inert materials filled with electrolyte [55,56]. Separators must be permeable to the ions and inert in the supercapacitor environment. Moreover, there are generally two current collectors (like copper foils, nickel foils, and gold foils) that are attached to the surface of electrodes to conduct the electrical current from electrodes. The electrodes, separator, and current collectors are immersed in an electrolyte, which enables the flow of ionic current between electrodes while preventing electronic current from discharging the capacitor, as illustrated in Fig. 1.

2.2. Energy storage mechanisms

The energy storage of supercapacitors is governed by the same principles as that of conventional capacitors. The energy storage is realized based on charging and discharging processes at the electrode–electrolyte interface. Thus, supercapacitors are characteristic with fast release and storage of energy. In comparison to conventional capacitors, the electrode of supercapacitors achieves a greater effective surface area, which contributes to a dramatical enhancement in capacitance by a factor of 10,000 than conventional capacitors [58]. Simultaneously, supercapacitors maintain low equivalent series resistance and high operational specific power. The energy storage mechanisms of supercapacitors include two kinds: (1) EDLC associated with adsorption of Coulombian charge near the electrode–electrolyte boundary; (2) pseudocapacitance originated from superficial redox reactions correlated to their respective potentials.

As mentioned above, the EDLC mechanism of supercapacitors is identical to that of conventional dielectric capacitors. For dielectric capacitors, the capacitance is related to the separation between the two charged plates, thus limiting the charge storage. Due to the significantly larger surface area of electrodes, supercapacitors can store much more energy [59]. As illustrated in Fig. 2a, the electrical charge is electrostatically stored due to reversible adsorption of ions of the electrolyte onto the electrode. The charge storage takes place directly across the double layer of the electrode material and there is no electrochemical reaction in the charging-discharging process. The capacitance arises owing to the true capacitance effect. With the help of the surface dissociation, ion adsorption from solution, and defects in the crystal lattice of electrodes, the surface charge is generated. The deficiency or excess of charge that is created on the electrode surface results in the formation of oppositely charged ions in the electrolyte near the electrode–electrolyte interface for electroneutrality. The thickness of the double layer is dependent on the electrolyte concentration and size of ions and is of order 5–10 Å for concentrated electrolytes. The most typical characteristic of EDLCs is the pure physical charge accumulation process occurring at the electrode/electrolyte interface. As a result, the capacitance of EDLCs tightly relies on the effective surface area and pore diameter of electrodes. Up to now, multifarious carbon materials (like activated carbon, carbon aerogels, carbon foams, and mesoporous carbon) with adjustable pore structure, good conductivity, and large surface area have been widely used as electrode materials for EDLCs. Although EDLCs have high power density and superb charging-discharging stability, they usually suffer from low energy density, thus promoting the appearance of pseudocapacitive materials.

2.3. Pseudocapacitance mechanism

Compared to EDLCs, pseudocapacitors that use conducting polymers and transition metal oxides/hydroxides/sulfides/nitrides usually deliver higher specific capacitances and energy density due to fast reversible redox reactions of faradaic nature enveloping the elevation of energy in pseudo-electrodes. Pseudocapacitance is non-electrostatic in nature and arises as a consequence of electrochemical charge-transfer accompanied by the finite amount of active material. In electrochemistry, the term pseudocapacitance designates a capacitive material of electrochemical nature with the linear reliance of charge stored on the width of potential window [61]. In pseudocapacitor, the capacitance is associated with the amount of charge accepted and varying potential.

Reactions like electrosorption or redox reactions of electroactive materials play a key role in the generation of a faradaic current in pseudocapacitor electrodes (Fig. 2b). The electrosorption occurs because of the chemisorption of electron donating anions (such as Cl-, B-, etc.). For the redox reactions, the charge exchange across the double layer happens to replace the static separation of charge. For supercapacitors, both the charge storage mechanisms (namely EDLC and pseudocapacitance) often co-exist. One storage mechanism becomes the dominant contributor to the specific capacitance while the another contributes very little. Pseudocapacitance-dominant electrode materials include metal oxides (like ruthenium oxides, manganese oxides, nickel oxides, cobalt oxides, copper oxides, and vanadium nitrides), conducting polymers (like polypyrrole (PPy) and polyaniline (PANI)), carbon-based hetero-atoms (such as N, P, or S-doped carbon), and nanoporous carbons with electroadsorbed hydrogen. Pseudocapacitive materials have high specific capacitance and energy density, but they usually suffer from low electrical conductivity and poor rate capability as well as inferior structural stability due to the repeated ion intercalation and de-intercalation.

3. Preparation, structure, and property of cellulose

In the nanostructure of plant cell walls, cellulose microfibers, composed of many cellulose nanofibrils, act as the skeleton component to combine with hemicellulose and lignin. The mixed structure contributes to the toughness and rigidity of cell walls. The interspace between cellulose microfibers and the space within the well-oriented vessels constitute the hierarchically porous nanostructure of plants, including macropores (>50 nm) and mesopores (2–50 nm). Cellulose microfibers can further be split into more sophisticated nanofibrils made up of elementary fibrils with smaller diameters (1.5–3.5 nm). The gap between these cellulose nanofibrils or elementary fibrils forms abundant micropores (less than2 nm). Cellulose has been paid much attention from various fields, such as supercapacitors [62], antimicrobial agents [63-65], nanogenerators [66] and negative permittivity materials [67].

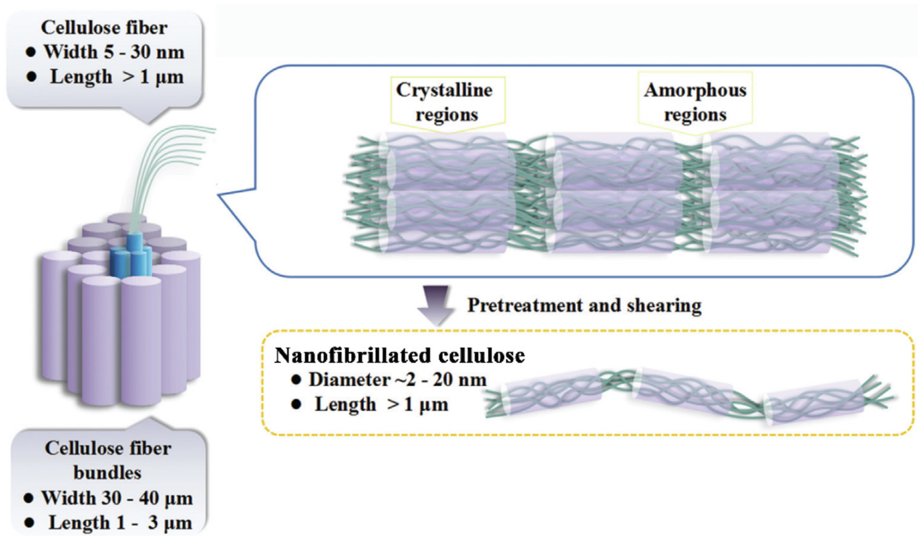

Fig. 4. Schematic diagram of nanofibrillated cellulose from cellulose fiber bundles (cellulose microfibers). (Reprinted with permission from [80]. Copyright 2021 The Royal Society of Chemistry.).

3.1. Cellulose microfibers

Cellulose microfibers (CMFs) are separated from biomass through the chemical purification combined with the mechanical isolation [68]. The sectional dimension of CMFs is from a few to tens of micrometers and the longitudinal dimension is several hundred micrometers. These fibers have rough surface and abundant active groups, which can intertwine through hydrogen bonds to assemble films with high mechanical strength by vacuum filtration. Nevertheless, the crystallinity of cellulose microfibers is usually lower than 60%, and thus water can easily penetrate into the amorphous region of microfibers, leading to the expansion of the fibers and the notable degradation of their mechanical property [69].

3.2. Regenerated cellulose

Native cellulose usually has cellulose I crystal structure. It is worth mentioning that cellulose I crystal structure includes two distinct crystal phases (Iα and Iβ). Cellulose Iα is widely found in the cell wall of some algae and in bacterial, whereas cellulose Iβ is generally found in tunicin and higher plants. In addition, Iα corresponds to a one-chain triclinic P1 unit cell (a = 6.717 Å, b = 5.9 62 Å, c = 10.400 Å, α = 118.08°, β = 114.80°, and γ = 80.37°), while Iβ corresponds to a two-chain monoclinic P21 unit cell (a = 7.784 Å, b = 8.201 Å, c = 10.38 Å, α = β = 90°, and γ = 96.5°). By contrast, due to the regeneration or mercerization treatments, the regenerated cellulose has cellulose II crystal structure, a monoclinic P21 phase (a = 8.10 Å, b = 9.03 Å, c = 10.31 Å, α = β = 90°, and γ = 117.10°) with two antiparallel chains in the unit cell (Fig. 3). Compared with cellulose I, cellulose II has some enhanced properties and stability that make it preferable for some textiles and pulp-derived materials [70]. Besides the differences in crystal structure, their mechanical strength is also different. The regeneration proceeds by putting the untreated, frozen or dried cellulose solution in a regeneration (or coagulating) bath to induce a phase separation of cellulose from soluble state to insoluble state. The kinetics of regeneration is mainly controlled by the relative velocity of the counter-diffusion process (i.e., solvent diffusion from the solution to the bath, and non-solvent diffusion from the bath into the solution) [71]. Especially, this solvent-exchange process results in a desolvation of cellulose molecules and reformation of the intra- and inter-molecular hydrogen bonds, which are beneficial for the formation of hydrogels/alcogels.

3.3. Nanofibrillated cellulose

Nanofibrillated cellulose has a large aspect ratio with length>2 μm and web-like entangled structure, leading to a high specific surface area [73]. Nanofibrillated cellulose from biomass cell walls is usually prepared through mechanical fibrillation process (Fig. 4). Several mechanical methods are available to prepare nanofibrillated cellulose such as high-pressure homogenization [74], high-intensity ultrasonication [75], microfluidization [76] and steam explosion [77]. In a typical mechanical process, the structure of cell walls and hydrogen bonds are destroyed by the high-speed shear and friction constantly without causing degradation on the cellulose, allowing the extraction of nanofibrillated cellulose with a diameter less 100 nm and length of 500 nm or longer. However, non-homogeneity and high energy-consumption are unavoidable for preparing nanofibrillated cellulose by mechanical methods. Chemical pretreatment before nanofibrillation is an efficient pathway to minimize energy requirement and improve fibrillation degree. A typical example is that Saito and Isogai [78] prepared completely individualized cellulose nanofibrils from wood cellulose fibers by a pioneering method of 2,2,6,6-tetrame thylpiperidine-1-oxyl radical (TEMPO) oxidation under moderate aqueous conditions. The C6 primary hydroxyls of cellulose can be converted to anionically charged C6 carboxylate groups by TEMPO-mediated oxidation, which forms strong electrostatic repulsion between nanocellulose in water and renders excellent dispersibility. However, the carboxylated nanofibrillated cellulose generally suffers from a low degree of polymerization and an inferior thermal stability. Enzymatic treatment of wood cellulose is a facile and green way, enabling a decrease in the required energy consumption for nanofibrillation process. Pääkko et al. [79] combine enzymatic hydrolysis with high-pressure homogenization and physic shearing to prepare nanofibrillated cellulose that mainly consist of fibrils with a diameter of ca. 5–6 nm and fibril aggregates (ca. 10–20 nm). In this enzyme pretreatment, the mono-component endoglucanase enzyme facilitates the cell wall delamination and prevents the blocking of the homogenizer even in small amount addition.

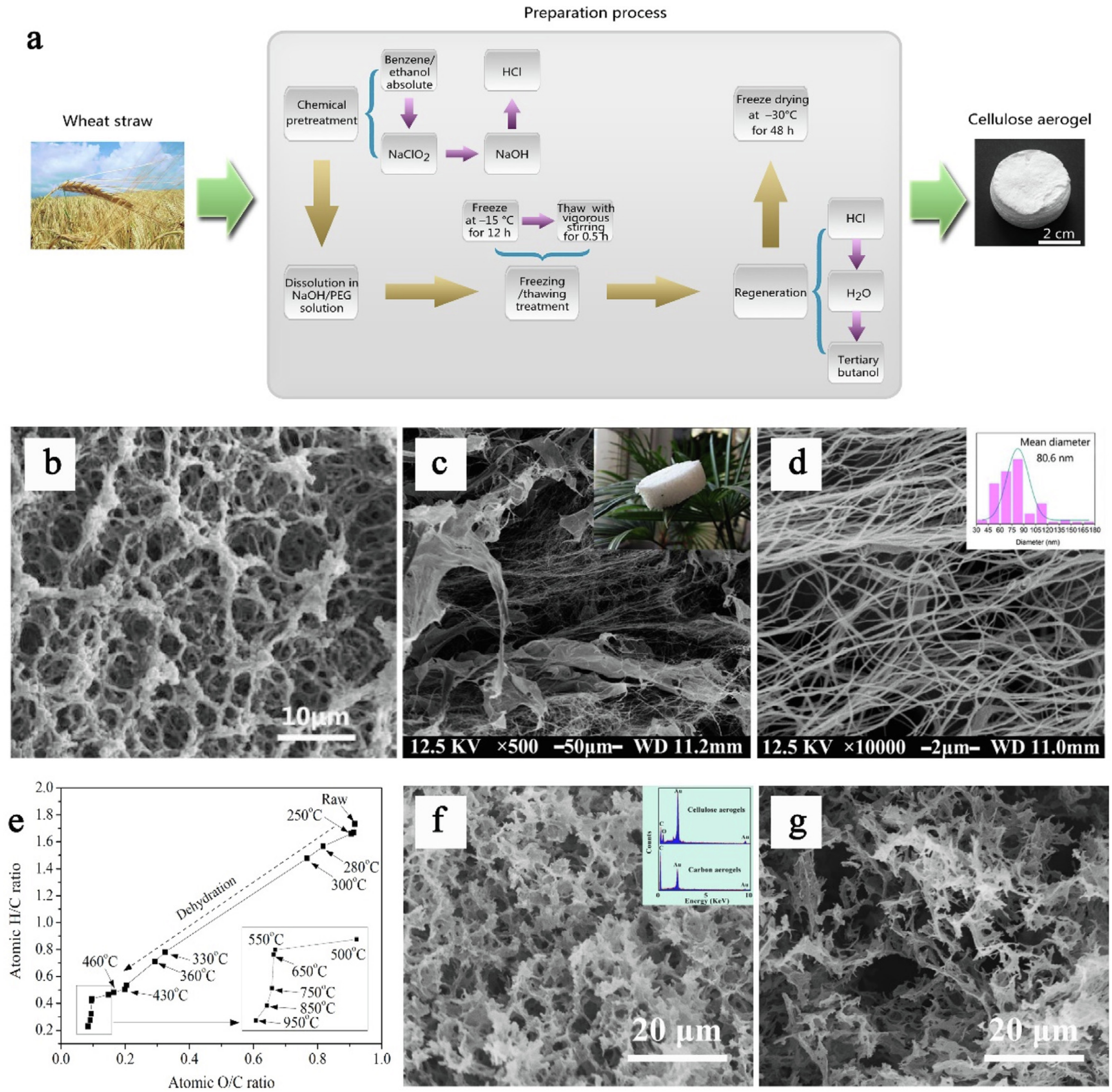

Fig. 5. Preparation method and microstructure of CDCAs and their precursors. (a) Schematic representation of RCAs prepared from wheat straw; (b) scanning electron microscopy (SEM) image of RCAs; (c-d) SEM images of NCAs (the inset in panel c is a digital photograph of the lightweight NCAs, and the inset in panel d presents the diameter distribution of the nanofibrillated cellulose); (e) Van Krevelan diagram of cellulose char from different temperatures; (f-g) SEM images of (f) RCAs and (g) CDCAs before and after pyrolysis (the inset in panel f shows the energy dispersive X-ray (EDX) spectra). ((a-b) Reprinted with permission from [44]. Copyright 2015 Taylor & Francis. (c-d) Reprinted with permission from [45]. Copyright 2015 Wiley. (e) Reprinted with permission from [85]. Copyright 2015 Elsevier. (f-g) Reprinted with permission from [79]. Copyright 2015 Elsevier.).

4. Preparation technologies of cellulose-derived carbon aerogels

Prior to the synthesis of CDCAs, acquiring their precursors (i.e., cellulose aerogels) is the prerequisite. There are two primary approaches for the preparation of cellulose aerogels based on the difference in the nano-disintegration of cellulose. The cellulose dissolution is an effective method to convert macroscopic cellulose to well-dispersed nano-sized cellulose. After the regeneration, the re-assembled cellulose is known as the regenerated cellulose. Due to the difficulty and complexity of cellulose dissolution in common water or organic solvents, the mechanical fibrillation is another powerful way to separate nanocellulose. The isolated cellulose is called as the nanofibrillated cellulose. The corresponding aerogels are identified as the regenerated cellulose aerogels (RCAs) and nanofibrillated cellulose aerogels (NCAs), respectively.

4.1. Preparation technology of regenerated cellulose aerogels

Cellulose dissolution is the core process of the preparation of RCAs. The hydrogen bond network, which is constructed by inter- and intra-hydrogen bonds within or between cellulose chains, makes cellulose feedstocks undissolved in common solvents. Besides the restriction of hydrogen bonds, the hydrophobic stacking resulted from theamphiphilic nature of cellulose also limits the cellulose dissolution [81]. A series of cellulose solvents emerge in response to the needs of times. From the perspective of organic chemistry, cellulose solvents can be categorized as "derivatizing" and "non-derivatizing" systems [82]. Historically, derivatizing solvents are firstly invented. For derivatizing solvents, the dissolution of cellulose is achieved by virtue of covalent modification to transform cellulose to an unstable ether, ester or acetal intermediate. These intermediates can then be treated to regenerate cellulose. For non-derivatizing solvents, cellulose dissolution occurs through intermolecular interactions. Based on differences in the compositions and action mechanisms, non-derivatizing solvents primarily include three classes: (1) aqueous complexing agents, (2) alkaline and related systems, and (3) non-aqueous systems. The purpose of non-derivatizing solvents is to decrease the self-interaction of cellulose chains and to hinder the hydrophobic association of cellulose.

After the dissolution, the cellulose solution is frozen to generate a 3D framework structure owing to the filling of ice crystals between cellulose components. Afterwards, the frozen cellulose solution is immersed in a coagulating bath containing anti-solvents. The introduction of anti-solvents is expected to interrupt the solubility equilibrium and to induce the re-assembly of cellulose molecules, leading to the occurrence of a solid- liquid phase separation. As a result, cellulose hydrogels are generated. With the help of special drying techniques (like freeze drying and supercritical drying) that minimize the drying stress, the liquid in hydrogels can be effectively removed and the corresponding cellulose aerogels are prepared (Fig. 5a). RCAs usually feature with high structural stiffness, strong mechanical strength, large surface area, and high porosity, due to the formation of developed 3D porous network structure (Fig. 5b).

4.2. Preparation technology of nanofibrillated cellulose aerogels

For the dissolution pathway to prepare cellulose aerogels, some cellulose solvents (for instance, cuprammonium hydroxide, cadoxen, and FeTNa) are eco-unfriendly and a part of solvents (e.g., ionic liquids) are expensive and require unique conditions when used (like anhydrous environment and low temperature). Regarding these limits, another interesting method (i.e., mechanical fibrillation) is developed to prepare nanocellulose suspensions. The aerogels originated from nanofibrillated cellulose, whose preparation process mainly depends on the isolation of nanofibrillated cellulose and freeze-drying treatment, show considerable merits from the perspective of environmental protection since no harmful solvents are required during the processing of nanofibrillated cellulose. Furthermore, some time-consuming necessary courses, such as dissolution and gelation related to RCAs, are also not involved in preparation of NCAs. Instead, some mild fast mechanical technologies, such as ultrasonic processing, high-pressure homogenization, and ball milling, are used for the isolation of nanofibrillated cellulose. Undoubtedly, the mechanical fibrillation demands for high energy consumption. In consequence, some pre-treatments like TEMPO oxidation and enzyme treatment are involved. For NCAs, the particular 3D slender network formed by the self-assembly of cellulose nanofibrils with diameters of a few nanometers to several tens of nanometers creates a large number of extraordinary performances in the density, adsorption, and flexibility [83]. As compared to RCAs, the diameter of the skeleton structure of NCAs is much smaller (Fig. 5c), thus leading to their higher flexibility and compressibility and lower density but inferior mechanical strength. Wan et al. prepared the NCA using coconut shell as feedstock through a chemical pretreatment-ultrasonic isolation-solvent exchange-freeze drying, achieves a low bulk density of 0.84 mg cm^{v3} and an ultrastrong oil-adsorption capacity (542 times that of the initial dry weight of diesel oil) after being modified by methyltrichlorosilane [45].

4.3. Conversion technology of cellulose aerogels to carbon aerogels

For the application in supercapacitors, it is of great importance to convert electrically insulating cellulose aerogels to electrically conductive carbon aerogels and to maintain the porous structure at the same time. The high-temperature pyrolysis treatment under the protection of inert gas (such as nitrogen and argon) is a universal method for both RCAs and NCAs to prepare carbon aerogels. Actually, the pyrolysis of RCAs and NCAs can be regarded as the pyrolysis of cellulose. For the pyrolysis of cellulose, there are primary five stages (Fig. 5d) [84]: (1) the initial temperature at which water evolved chemically from cellulose pyrolysis is ca. 200 °C; (2) the cleavage of intra- and inter-molecular hydrogen bonds (H-bonds) and the subsequent dehydration are the main reactions as temperature below 300 °C, among which intra-molecular dehydration is the predominated reaction; (3) dehydration occ-urs principally inter- molecularly along with decarbonylation, ring-opening, and aromatization over 300 °C; (4) dehydration is almost accomplished at 430 °C and the smallest aromatic clusters grafted with oxygenated groups undergo notable deoxygenation and condensation in the range of 430–650 °C; (5) dominant reaction shifts from deoxygenation to dehydrogenation as the temperature exceeds 650 °C and the char is highly aromatic with large aromatic systems made up of over six fused ring structures. Usually, the pyrolysis temperature of cellulose aerogels is above 800 °C for the sake of

Table 1
Comparison of electrochemical performances of some typical CDCA-based supercapacitor electrodes.

Electrodes	Specific capacitance	Cycle stability	Energy density of devices	Origins	Refs.
KOH-activated CDCAs	381F g^{-1} (at 5 mV s^{-1})	–	–	RCAs	86
CO_2-activated CDCAs	302F g^{-1} (at 0.5 A g^{-1})	93% (4000 cycles)	–	RCAs	87
N, S dual-doped CDCAs	329F g^{-1} (at 0.5 A g^{-1})	90% (5000 cycles)	10.3 W h kg^{-1} (at 130 W kg^{-1})	RCAs	89
N-doped CDCAs	185F g^{-1} (at 1.0 A g^{-1})	–	–	RCAs	90
PPy/CDCAs	387.6F g^{-1} (at 0.5 A g^{-1})	92.6% (10000 cycles)	–	RCAs	97
FeOCl@CDCAs	647F g^{-1} (at 0.8 A g^{-1})	91% (10000 cycles)	289 mW h cm^{-2} (at 1.8 mW cm^{-2})	RCAs	57
Interconnected nanofiber-based CDCAs	235F g^{-1} (at 0.5 A g^{-1})	–	23.05 W h kg^{-1} (at 213 W kg^{-1})	NCAs	98
Wood-based CDCAs	140F g^{-1} (at 0.5 A g^{-1})	–	–	NCAs	61
CDCAs with a N-containing sandwich-like wall	274F g^{-1} (at 1.0 A g^{-1})	97.7% (4000 cycles)	8.51 W h kg^{-1} (at 500.18 W kg^{-1})	NCAs	99
RGO/CDCAs	398.47F g^{-1} (at 0.5 A g^{-1})	99.77% (10000 cycles)	–	NCAs	100
MnO_2/CDCAs	171.1F g^{-1} (at 0.5 A g^{-1})	98.4% (5000 cycles)	8.6 W h kg^{-1} (at 619.2 W kg^{-1})	NCAs	101

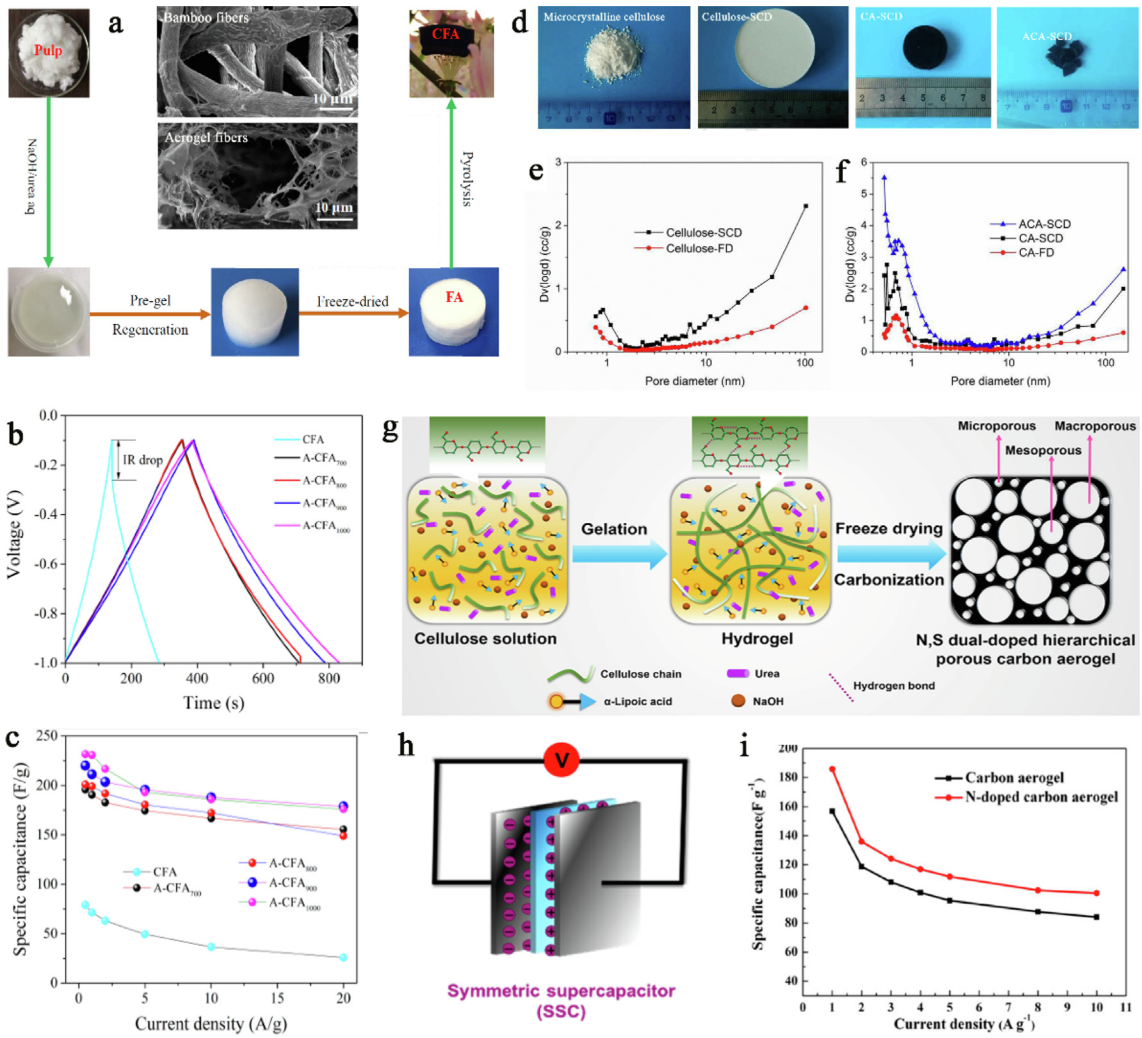

Fig. 6. Preparation method, microstructure, and energy storage property of RCA-derived CDCAs. (a) Schematic illustration of the fabrication process of the carbon fiber aerogel (CFA) and its microstructure; (b) galvanostatic charge/discharge (GCD) curves of the activated CFA (A- CFAx, X means the activated temperature) and CFA electrodes at current density of 0.5 A g^{-1}; (c) capacitance retention of the A-CFAx at different current densities; (d) photographs of typical samples including microcrystalline cellulose powders, supercritical CO2-dried aerogel (cellulose- SCD), carbon aerogel (CA-SCD), and activated carbon aerogel (ACA-SCD); (d-e) pore-size distributions of typical cellulose aerogels and carbon aerogels; (g) schematic fabrication process of the N, S dual-doped hierarchical porous carbon aerogel; (h) schematic SSCs; (i) specific capacitances of the carbon aerogel and N-doped carbon aerogel at different currents. ((a-c) Reprinted with permission from [86]. Copyright 2018 Elsevier. (d-f) Reprinted with permission from [87]. Copyright 2016 Elsevier. (g-h) Reprinted with permission from [89]. Copyright 2020 American Chemical Society. (i) Reprinted with permission from [90]. Copyright 2019 MDPI.).

complete conversion. Wan et al. [79] demonstrates that the RCA-derived carbon aerogel inherits the 3D interconnected network from the RCA after the pyrolysis (Fig. 5d and e) and receive new functions including hydrophobicity, conductivity, and flame retardance.

5. Cellulose-derived carbon aerogels-based supercapacitor electrodes

Because of the difference in the physical structure between RCAs and NCAs, there are notable differences in the specific surface area, pore diameter distribution, flexibility, and mechanical strength, which play significant roles in determining their application forms. For example, fragile carbon aerogels are commonly utilized in powder form and mixed with the conductive agent (such as acetylene black) and adhesion agent (such as polyvinylidene fluoride, PVDF). The mixture is paint on the surface of nickel foam and pressed into a nickel foam-supported electrode. By comparison, flexible carbon aerogels can be directly employed as the substrate for free-standing electrodes (Table 1), which avoids the introduction of adhesion agents.

5.1. RCA-derived carbon aerogels

5.1.1. Direct application as porous electrodes

The high entanglement density in the preparation of RCAs facilitates the formation of large specific surface area and high porosity. Thus, the application of RCA-derived carbon aerogels in supercapacitors is similar to that of activated carbon. Yang et al. [86] used NaOH/urea aqueous solution to dissolve bamboo cellulose fibers for the fabrication of porous RCAs (Fig. 6a). After the pyrolysis and KOH activation (to increase specific surface area), the activated RCA-derived carbon aerogel is mixed with carbon black and PVDF at a weight ratio of 90:5:5 and fixed roughly between two pieces of nickel foam under a pressure of 10 MPa. The nickel foam- supported carbon aerogel electrode exhibits a high specific capacitance of 381F g^{-1} in aqueous electrolyte of 6 M KOH, which is improved by 150% compared to unactivated sample (Fig. 6b and

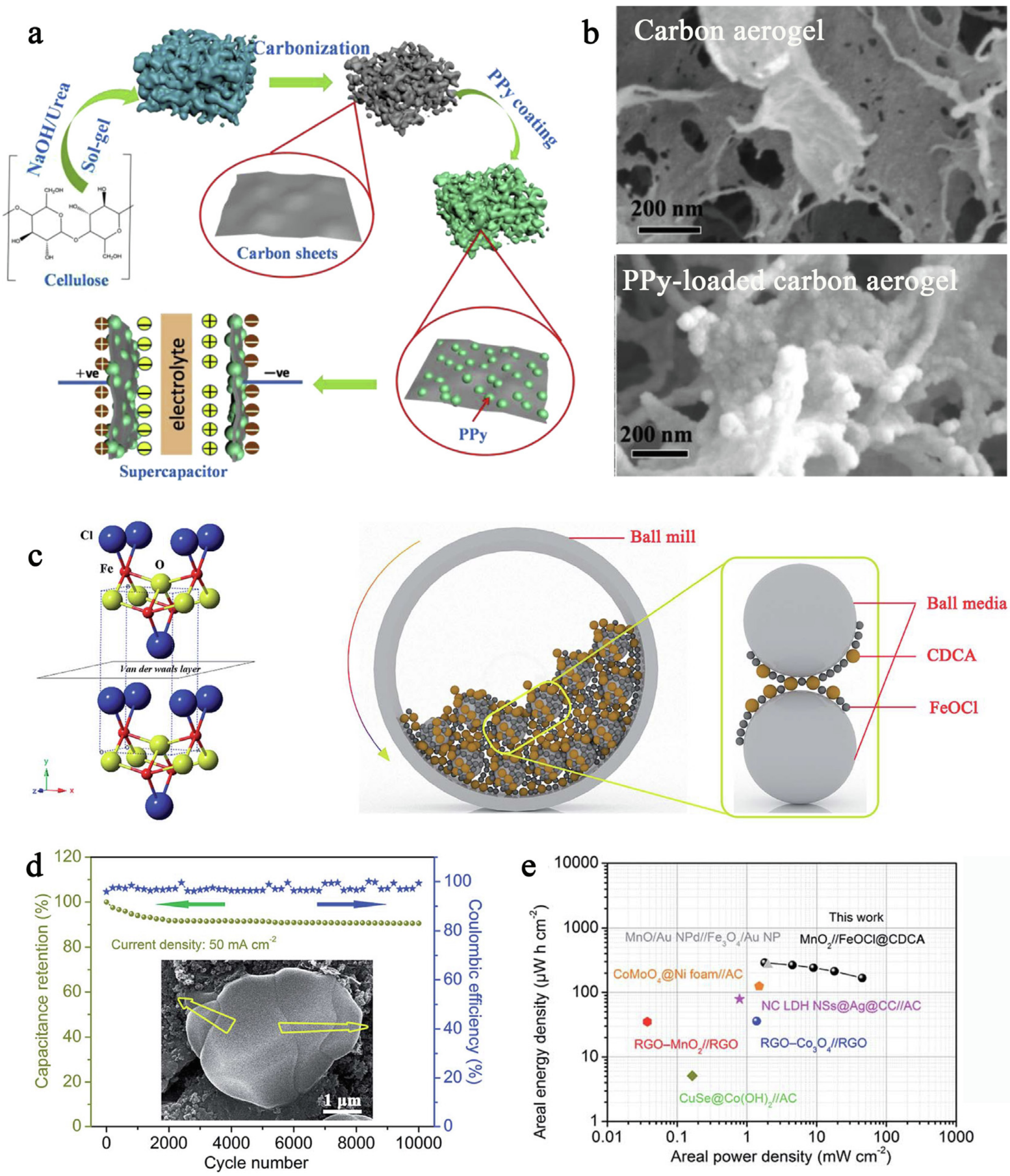

Fig. 7. Preparation method, microstructure, and energy storage property of RCA-derived CDCA-based composite electrodes. (a) Schematic fabrication process of the 3D hierarchical porous carbon aerogel/PPy composite; (b) SEM images of the carbon aerogel and carbon aerogel/PPy composite; (c) schematic for the crystalline structure of FeOCl and the fabrication of the FeOCl@CDCA by ball-milling; (d) cycle stability and coulombic efficiency of the FeOCl@CDCA at 50 mA cm^{-2} and the inset is its SEM image after 10,000 cycles; (e) Ragone plot of the MnO2//FeOCl@CDCA ASC compared with the data of some other ASCs. ((a-b) Reprinted with permission from [96]. Copyright 2019 Elsevier. (c-e) Reprinted with permission from [57]. Copyright 2019 The Royal Society of Chemistry.).

c). The reasonable micropores and mesopores contribute to a superb retention rate of 90% at high scan rate of 200 mV s^{-1}. The low equivalent series resistance and small charge transfer resistance in the carbon aerogel also suggests an inspiring capacitor behavior. Zu et al. [87] prepared a high-surface-area carbon aerogel by pyrolysis of a RCA that is prepared by dissolving microcrystalline cellulose in NaOH solutions followed by gelation, regeneration and supercritical CO2 drying (Fig. 6d). The obtained carbon aerogel is further treated via a CO2 activation process. The RCA dried by supercritical CO2 shows a high specific surface area (299 m^2 g^{-1}) and the obtained carbon aerogel preserves the interconnected 3D nanostructure and has a high specific surface area of 892 m^2 g^{-1}, which increases to 1873 m^2 g^{-1} after CO2 activation. The CO2 activation produces abundant nanopores (Fig. 6e and f). The highly porous and interconnected nanostructure provides efficient migration of electrolyte ions and electrons. The resultant activated carbon aerogel shows the specific capacitances of 302 and 205F g^{-1} at the current density of 0.5 and 20 A g^{-1}, respectively, indicative of a good rate capability. It is found that these above activation treatments contribute to the enhancement of surface area, thus improving the energy storage ability. For pure carbon materials, there is only the EDLC mechanism proceeding energy storage. Nevertheless, the incomplete pyrolysis reserves a small number of oxygen-containing groups in CDCAs. As a result, faradaic reactions occurring on the surface of CDCAs, such as reaction between the carbonyl (C = O) and hydroxyl (C–OH) groups (i.e., C = O + H^+ + e^- ↔ C–OH), may produce the pseudocapacitance [88].

5.1.2. Doping and combination as hybrid electrodes

To further improve the energy storage ability of CDCAs, some effective methods, like hetero-atom doping and integration with pseudocapacitive materials, have been well developed [33,62,91-95]. Dang et al. [89] dissolved α-lipoic acid with disulfide bond and carboxyl group in the NaOH/urea solution. The ternary solvent is used to dissolve cellulose. Followed by gelling and carbonization, the N, S dual-doped hierarchical porous carbon aerogel is obtained (Fig. 6g). Because the fabricated carbon material has a proper structure and a uniform heteroatom doping, its capacitance can reach 329F g^{-1} at 0.5 A g^{-1}, 1647.5 mF cm^{-2} at 2.5 mA cm^{-2}, and the fine rate property is 215F g^{-1} at 10 Ag^{-1} and 1075 mF cm^{-2} at 50 mA cm^{-2}, respectively. A symmetric supercapacitor (SSC) assembled by the carbon aerogels (as illustrated in Fig. 6h) yields a high energy density of 10.3 W h kg^{-1} at a power density of 130 W kg^{-1}. Recently, by virtue of the porous structure and strong adsorption ability of cellulose aerogels, it is interesting and fascinating for a dye adsorption method to achieve hetero-atom doping. For instance, E et al. [90] used an alkaline peroxide mechanical pulp (APMP) fiber aerogel as the sorbent and adopted the aerogel saturated with Rhodamine B (RhB) dye to obtain N-doped carbon aerogels for supercapacitors. The RhB adsorbed onto the APMP aerogel is used as a nitrogen source, and the N-doped carbon aerogel is obtained after carbonization of the aerogel saturated with RhB dye. After the doping, a 19.4% of improvement in the specific capacitance is achieved at the current density of 1 A g^{-1} (Fig. 6i). These electrochemical properties suggest the role of hetero-atom doping in improving the energy storage ability of carbon electrode materials, especially based on renewable and low-cost cellulose.

In contrast to the hetero-atom doping, integrating CDCAs with pseudocapacitive materials (e.g., conductive polymers and metallic compounds) is usually more powerful. A typical example is that

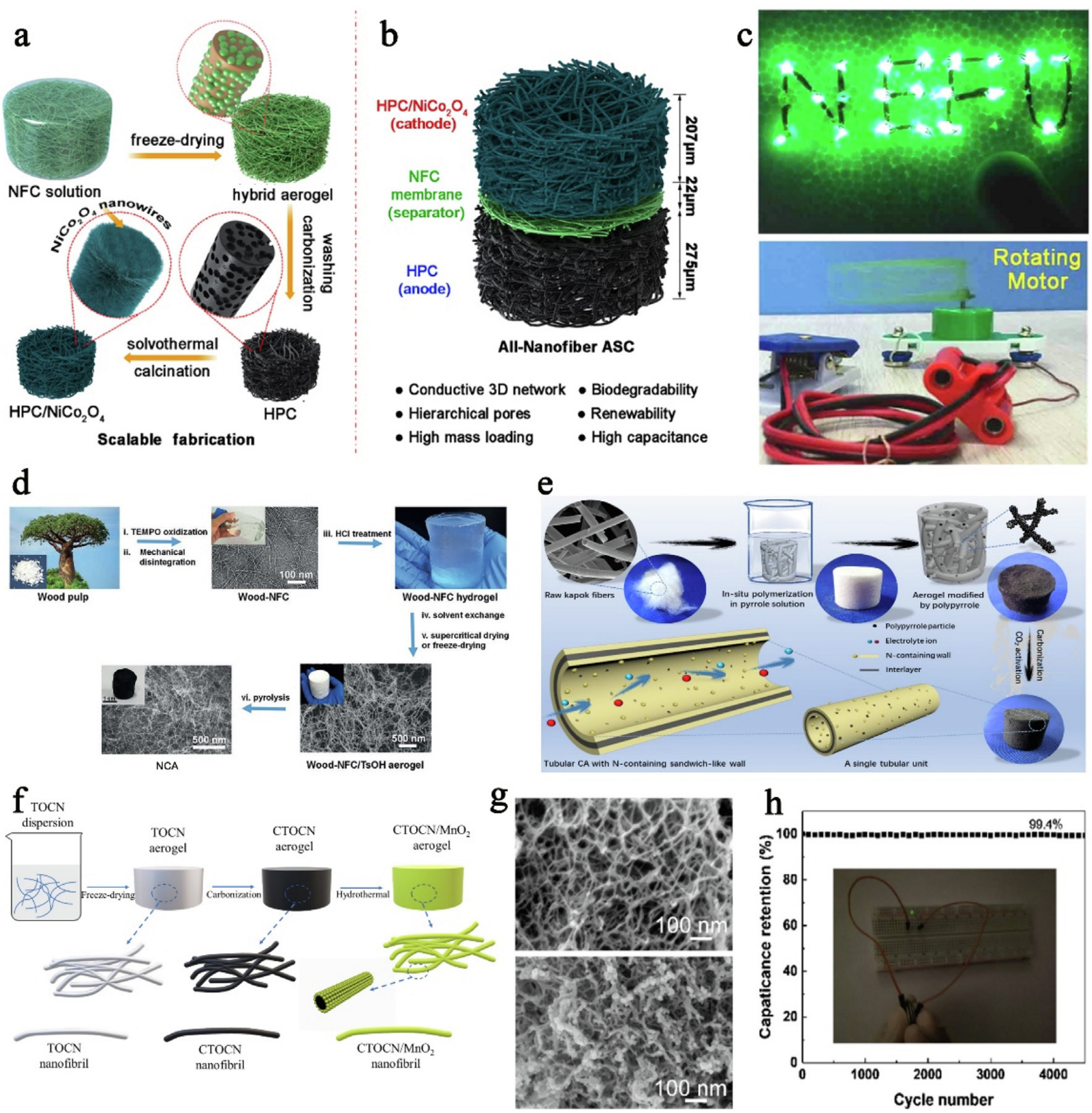

Fig. 8. Preparation method, microstructure, and energy storage property of NCA-derived CDCAs. (a-b) Schematic illustrations of the (a) fabrication of nanocellulose-derived hierarchical porous carbon (HPC) and HPC/NiCo2O4 composite aerogels and (b) their assembly into an all- nanofiber ASC device; (c) optical photographs of the green LEDs and a fan powered by a prototype ASC cell; (d) illustration of the fabrication of wood-derived NCAs; (e) synthetic route to the tubular carbon aerogel with N-containing sandwich-like wall; (f) schematic illustration for the synthesis of the carbon nanofibril (CTOCN)/MnO2 composite aerogel; (g) SEM images of the CTOCN (up) and CTOCN/MnO2 (down); (h) cycling stability of the CTOCN/MnO2//activated carbon ASC at 3 A g^{-1} over 4500 cycles. ((a-c) Reprinted with permission from [97]. Copyright 2019 American Chemical Society. (d) Reprinted with permission from [61]. Copyright 2018 Wiley. (e) Reprinted with permission from [98]. Copyright 2019 Elsevier. (f-h) Reprinted with permission from [100]. Copyright 2021 Elsevier.)

Zhuo et al. [96] used cellulose as a carbon precursor to prepare a porous carbon aerogel as a support for conductive polymer PPy (Fig. 7a). The hierarchical porous structure not only enables the efficient penetration and homogeneous loading of PPy throughout the carbon network (Fig. 7b), but also ensures a rapid transfer of electrolytes and the high accessibility of PPy. The as-prepared hybrid shows a high specific capacitance of 387.6 F g^{-1} (0.5 A g^{-1} in 1.0 M H2SO4) and superb cycling stability (92.6% capacitance retention after 10,000 cycles). Wan et al. [57] introduced a highly conductive and porous CDCA substrate to combine with orthorhombic FeOCl with a self-stacked laminated structure (Fig. 7c). The mechanical stability and charge-storage kinetics of FeOCl are significantly enhanced due to the presence of CDCA. FeOCl@CDCA delivers an ultra-high areal specific capacitance of 1618 mF cm^{-2} (647F g^{-1}) at 2 mA cm^{-2} and superb cycle stability with less than 10% capacitance loss after 10,000 cycles (Fig. 7d). An asymmetric supercapacitor (ASC), fabricated using a FeOCl@CDCA anode and a MnO2 cathode, displays a highly competitive energy/power density (289 mW h cm^{-2} at 1.8 mW cm^{-2}) (Fig. 7d) and excellent rate capability and cycle stability.

5.2. NCA-derived carbon aerogels

5.2.1. Direct application as free-standing electrodes

Distinguished with the mechanically strong and robust skeleton of RCAs, NCAs are composed of a threadlike fiber network and commonly exhibit a superior flexibility. As a result, NCAs can be directly utilized as self-supported electrodes or act as free-standing substrates to combine with other types of electrochemically active substances. Zhang et al. [97] demonstrated a high-performance all-nanofiber ASC assembled by using a NCA-derived porous carbon aerogel anode, a mesoporous nanocellulose membrane separator, and a NiCo2O4 cathode with a NCA-derived carbon aerogel as the support matrix (Fig. 8a and b). The carbon aerogel possesses a porous architecture comprising interconnected nanofibers with an ultrahigh specific surface area of 2046 m^2 g^{-1}. When integrated with the mesoporous feature of the nanocellulose membrane separator, these properties facilitate the quick delivery of ions and electrons even with a thick (up to several hundreds of micrometers) and highly loaded (5.8 mg cm^{-2}) ASC design. The all-nanofiber ASC demonstrates a high electrochemical property (64.83F g^{-1} and 10.84F cm^{-3} at 0.25 A g^{-1}, 32.78F g^{-1} and 5.48F cm^{-3} at 4 A g^{-1}) and is able to power a light-emitting diode (LED) or drive a fan rotation (Fig. 8c). Another typical case is that Li et al. [61] reported a catalytic pyrolysis process to fabricate ultrathin carbon nanofiber aerogels from the wood-based NCAs (Fig. 8d), which guarantees high carbon residual and well maintenance of the nanofibrous structure during the thermal decomposition of NCAs. The wood-derived carbon nanofiber aerogels exhibit excellent electrical conductivity, a large surface area, and potential as a binder-free electrode material for supercapacitors. Particularly, the carbon fiber aerogel electrodes possess a specific capacitance of 140F g^{-1} at 0.5 A g^{-1} and maintain a high capacitance of 90F g^{-1} at 20 A g^{-1}, corresponding to a good capacitance retention ratio of 64%. Sun et al. [98] used raw kapok fibers containing tubular structures as feedstocks to create a kapok aerogel with PPy shell (Fig. 8e). After the pyrolysis, an ultralight carbon aerogel with a N-containing sandwich-like wall (made up of pyrrolic/pyridine nitrogen, quaternary nitrogen/oxidized nitrogen, and pyridinic nitrogen) yields a 72F g^{-1} increase in the capacitance compared with that of a cattail carbon aerogel without a tubular structure. Specifically, the optimal sample has a specific capacitance of 274F g^{-1} at a current density of 1 A g^{-1} with an energy density of 8.51 Wh kg^{-1}, a superior rate capability, and superior cycling stability with a capacitance retention of 97.7% after 4000 cycles. All these NCA-derived carbon aerogels work in the form of self-supported electrodes in supercapacitors without any conductive and adhesion agents.

5.2.2. Substrates for hybrid electrodes

By introducing electrochemically active substances to the nanofibrillated cellulose suspension liquid and then undergoing freeze drying and high-temperature treatments, the NCA-based composite aerogels are fabricated. The synergistic effect from various active materials is expected to improve the enrgy storage ability. Typically, Yang et al. [99] reported an eco-friendly and highly scalable supercapacitor electrode by fabricating 3D carbon-based aerogels using TEMPO- oxidized cellulose nanofibrils (TOCN) and graphene oxide (GO) as precursors via a facile ion-exchange method followed by freeze-drying. The obtained TOCN/GO aerogels are further carbonized in argon atmosphere to prepare the carbon nanofibrils with improved conductivity and transform GO to reduced graphene oxide (RGO). The electrochemical performance of the carbon-based aerogels achieves a high specific capacitance of 398.47F g^{-1} at the current density of 0.5 A g^{-1}. Moreover, the initial capacitance keeps 99.77% after 10,000 charging/discharging cycles (Fig. 8f), revealing the outstanding cycling stability.

In addition to integrating cellulose suspension liquid with electrochemically active substances, the pyrolytic carbon aerogel is also a crucial target for the combination and medication. Chen et al. [100] carbonized the TEMPO-oxide cellulose nanofibrils (TOCN) to fabricate carbon nanofibrils. Carbon nanofibril/MnO2 hybrid aerogels are then obtained with the help of a fast hydrothermal method (Fig. 8f), through which MnO2 is strongly attached to the surface of carbon nanofibrils (Fig. 8g). The self-supported aerogel electrode has a maximum specific capacitance of 171.1F g^{-1} at 0.5 A g^{-1} and can still remain 98.4% of the original value after 5000 cycles at 3 A g^{-1}. Moreover, an ASC using the hybrid aerogel as the positive electrode and activated carbon (AC) as the negative electrode (Fig. 8h) delivers a high energy density of 8.6 W h kg^{-1} at a power density of 619.2 W kg^{-1} an excellent cycle life with 99.4% retention of the first cycle after 4500 cycles at 3 A g^{-1}.

6. Conclusion and prospects

Thanks to superior electrical conductivity, controllable porosity, good mechanical property, and excellent environmental friendless, CDCAs have become a green porous platform for the development of supercapacitor electrodes. CDCAs are fabricated by pyrolyzing cellulose aerogel precursors under the protection of inert gas. The preparation of cellulose aerogels can be mainly divided into two ways: (1) dissolution-regeneration- freeze or supercritical drying; and (2) mechanical fibrillation-solvent replacement-freeze or supercritical drying. The resultant cellulose aerogels are known as RCAs and NCAs, respectively. The former, featured with high porosity and strong structural stiffness, is commonly mixed with conductive and adhesion agents and then pressed onto a current collector. By contrast, the latter has more superior flexibility and thus commonly can directly act as the substrate of free-standing electrodes. Besides the direct use of CDCAs in supercapacitor electrodes, there are two important routes for the improvement of energy storage ability, i.e., hetero-atom doping and integration with pseudocapacitive materials. Unquestionably, CDCAs open up a new promising avenue to create enticing materials with desired electrochemical performances. This review attempts to provide an extensive vision of preparation, property, and supercapacitor applications of CDCAs. Nevertheless, there are still some challenges and research opportunities that need to be paid more attention in the future:

(1) Optimizing the preparation technique of cellulose aerogel precursors. Up to now, some procedures in the preparation of cellulose aerogels, e.g., the regeneration of cellulose solution, are time-consuming. A part of processes like mechanical fibrillation are quite energy-consuming. Thus, it is of great significance to optimize the preparation technique of cellulose aerogel precursors.

(2) Large-scale manufacturing. The synthesis of CDCA supercapacitor electrodes has flourished over the past decade, but the fabrication is still limited to the laboratory scale. It is still difficult to perform large-scale production of NCAs or RCAs with tunable sizes because of the lack of relevant equipment. Thus, developing effective corollary equipment will be beneficial for the large-scale manufacturing.

(3) Upgradation of ancillary materials and devices. For supercapacitors, CDCA-based electrodes need to be connected to other components like electrolytes, packaging materials, and current collectors. Nevertheless, it is still challenging to achieve devices with customizable size and weight. Also, the fabrication techniques and protective and packaging techniques should be upgraded.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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